Course Code	21MAB201T	Course Name	TRANSFORMS AND BOUNDARY VALUE PROBLEMS	Course Category	B	BASIC SCIENCES	L	T	P	C
							3	1	0	4

Pre-requisite Courses	*Nil*	Co- requisite Courses	*Nil*	Progressive Courses	*Nil*
Course Offering Department	*Mathematics*	Data Book / Codes / Standards		*Nil*	

Course Learning Rationale (CLR):	*The purpose of learning this course is to:*
CLR-1:	*analyze partial differential equations, and interpret the solutions related to PDE in engineering problems*
CLR-2:	*compute the Fourier series expansion and express the sine and cosine series*
CLR-3:	*analyze one-dimensional wave and heat equations using PDE and Fourier series concepts*
CLR-4:	*analyze Fourier transforms and their properties*
CLR-5:	*analyze Z transform for solving discrete-time Signal problems*

Course Outcomes (CO):	*At the end of this course, learners will be able to:*	Program Outcomes (PO)												Program Specific Outcomes		
		1	2	3	4	5	6	7	8	9	10	11	12			
		Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning	PSO-1	PSO-2	PSO-3
CO-1:	*construct and solve partial differential equations using various techniques*	*3*	*3*	-	-	-	-	-	-	-	-	-	-	-	-	-
CO-2:	*explain the Fourier series expansion of a function in terms of sine and cosine series*	*3*	*3*	-	-	-	-	-	-	-	-	-	-	-	-	-
CO-3:	*identify partial differential equations and utilize Fourier series techniques to solve one dimensional wave and heat equations*	*3*	*3*	-	-	-	-	-	-	-	-	-	-	-	-	-
CO-4:	*apply Fourier transforms techniques in signal analysis*	*3*	*3*	-	-	-	-	-	-	-	-	-	-	-	-	-
CO-5:	*solve discrete-time signal problems using Z transforms*	*3*	*3*	-	-	-	-	-	-	-	-	-	-	-	-	-

Unit-1 - Partial Differential Equations	***12 Hour***
Formation of partial differential equations by eliminating arbitrary constants & arbitrary functions- Solutions of standard types of first order partial differential equations – Lagrange's linear equation – Linear partial differential equations of second and higher order with constant coefficients of homogeneous types.	
Unit-2 - Fourier Series	***12 Hour***
Dirichlet's conditions – General Fourier series – Odd and even functions - Half range sine and cosine series - Parseval's identity – Harmonic Analysis	
Unit-3 - Applications of Partial Differential Equations	***12 Hour***
Classification of second order partial differential equations - Method of separation of variables – Solutions of one dimensional wave equation - One dimensional equation of heat conduction (Insulated edges excluded) - Steady state condition with zero boundary - Steady state condition with non-zero boundary conditions	
Unit-4 - Fourier Transforms	***12 Hour***
Fourier transform pair – Properties -Fourier sine and cosine transforms – Properties– Transforms of simple functions - Convolution theorem (without proof) – Parseval's identity.	
Unit-5 – Transforms	***12 Hour***
Z - transforms – Properties of Z transforms – Inverse Z transforms – Convolution theorem (without Proof) – Solution of linear difference equations with constant coefficients using Z-transform	

Learning Resources	1. *Erwin kreyszig, Advanced Engineering Mathematics, 10th Edition, John Wiley & Sons, 2015.* 2. *B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 43rd Edition, 2015.* 3. *Veerarajan T., Transforms and Partial Differential Equations, Tata McGraw-Hill, New Delhi, 3rd edition,2012.Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 2010 3rd Edition*	4. *N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, New Delhi, 10th edition,2016.* 5. *Kandasamy P., etal. Engineering Mathematics, Vol.II & Vol.III (4th revised edition), S. Chand & Co., New Delhi,2000*

Learning Assessment

	Bloom's Level of Thinking	*Continuous Learning Assessment (CLA)*				*Summative Final Examination (40% weightage)*	
		Formative CLA-1 Average of unit test (50%)		*Life-Long Learning CLA-2 (10%)*			
		Theory	*Practice*	*Theory*	*Practice*	*Theory*	*Practice*
Level 1	*Remember*	*20%*	-	*20%*	-	*20%*	-
Level 2	*Understand*	*20%*	-	*20%*	-	*20%*	-
Level 3	*Apply*	*30%*	-	*30%*	-	*30%*	-
Level 4	*Analyze*	*30%*	-	*30%*	-	*30%*	-
Level 5	*Evaluate*	-	-	-	-	-	-
Level 6	*Create*	-	-	-	-	-	-
	Total	*100 %*		*100 %*		*100 %*	

Course Designers

Experts from Industry	**Experts from Higher Technical Institutions**	**Internal Experts**
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